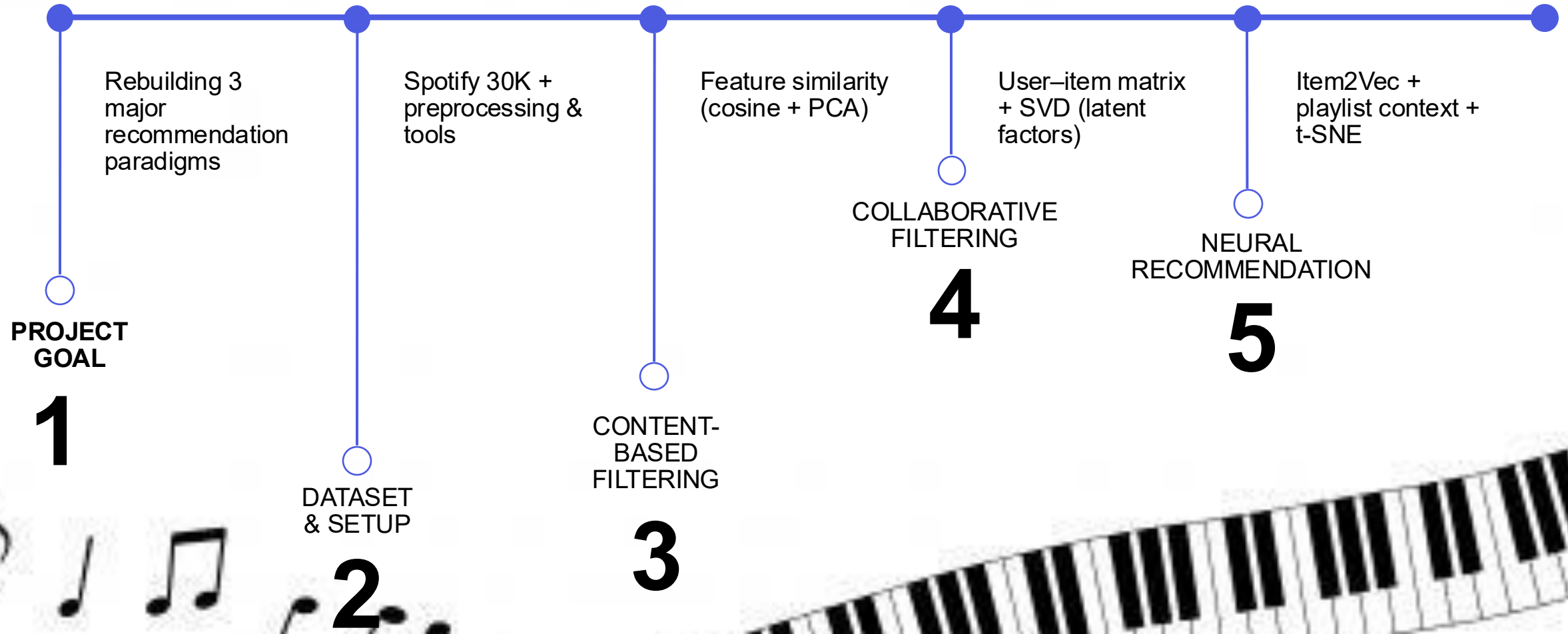


RECONSTRUCTING THE HISTORY OF MUSIC RECOMMENDATION



INTRODUCTION & MOTIVATION



Introduction

The explosion of digital music platforms created an overwhelming amount of content.

Users rely on recommendation systems to navigate millions of tracks.

Understanding *how* these systems evolved helps us design better, more transparent models.



Motivation

Shift: rules $\rightarrow$ statistics $\rightarrow$ neural models

Need to understand their logic & limits

Build transparent, comparable systems



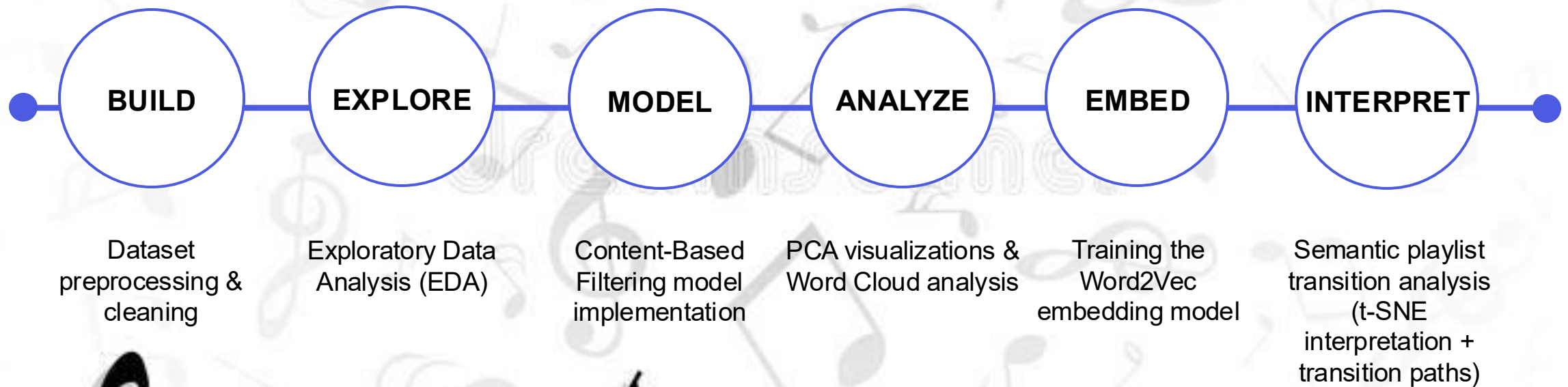
Objectives

Reconstruct the three recommendation paradigms.

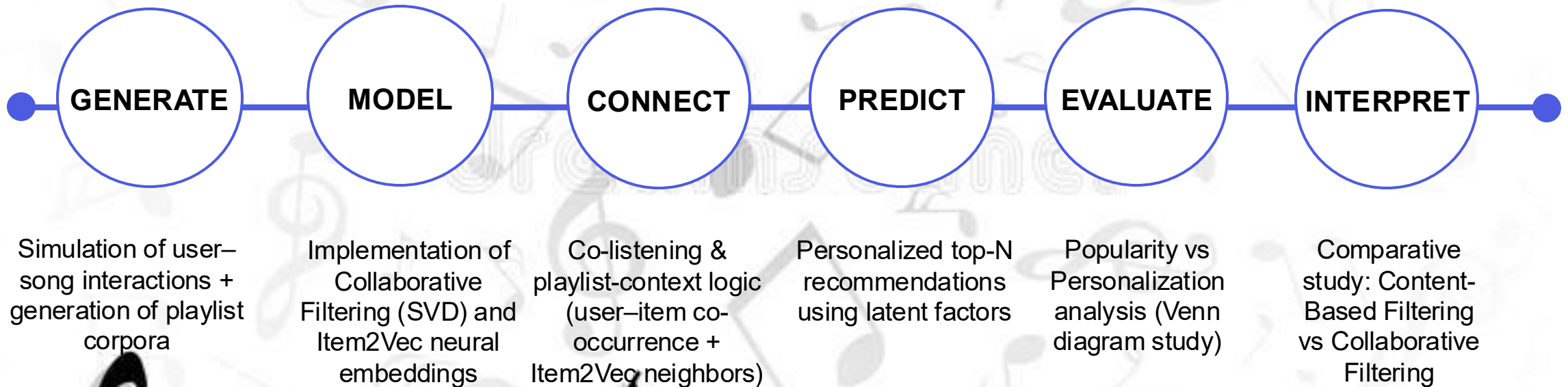
Analyze their behavior under the same dataset.

Identify what each method captures uniquely.

Contribution – Andra Mihaela Andruță



Contribution – Ioana Alexandra Tunaru



DATASET OVERVIEW

Dataset Summary

- Dataset extracted from [Kaggle's Spotify Tracks Dataset](#).
- Contains 29,865 unique songs after cleaning.
- Each entry includes track name, artist, and playlist-based genre.
- Includes 7 audio features describing musical characteristics.

Data Cleaning

- Removed missing values and duplicate tracks.
- Kept 10 essential columns used in our experiments.
- Standardized genre labels for consistency.
- Verified feature distributions before modeling.
- Final dataset prepared for all three recommendation paradigms.

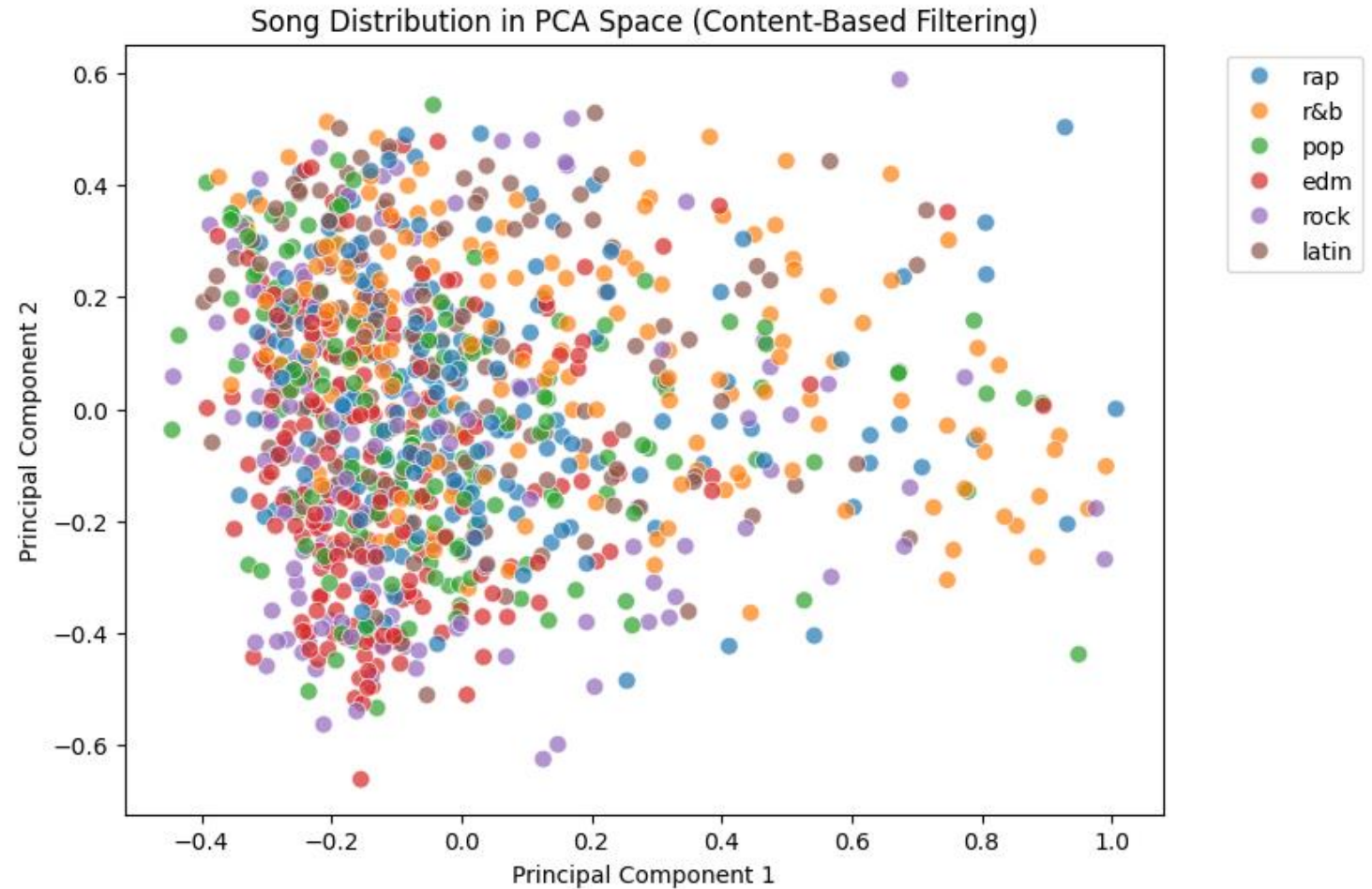
Feature Normalization

- Most Spotify audio features are already scaled 0–1.
- All 7 columns normalized using Min–Max scaling.
- Ensures fair comparison between all features.
- Normalized matrix used in similarity scores, SVD, and Word2Vec.
- Ready for content-based, collaborative, and embedding models.

CONTENT-BASED FILTERING (CBF)

- Uses **audio features** (danceability, energy, valence, tempo, acousticness, liveness, speechiness).
- Computes similarity using **cosine distance**.
- Recommends songs with **similar sound** characteristics.
- Does not require user interaction history.

This PCA plot shows how **songs cluster based on audio features**. Tracks with similar sound profiles appear closer together, supporting the idea behind content-based filtering.



COLLABORATIVE FILTERING (SVD)

- Learns from **collective user behavior** instead of audio features.
- Uses User–Item interactions to detect similar users or co-listened songs.
- Applies Matrix Factorization (SVD) to learn **latent taste factors**.
- Reconstructs missing ratings and predicts user preferences.
- **Co-listening logic**: songs frequently played together are considered similar.

SVD recommends new songs by predicting which tracks best match the user's **hidden taste profile**.

=== SMART-SVD Recommendations for user_id = 15 ===

Songs already listened to (with rating):

	track_name	track_artist	playlist_genre	rating
2	Only God Can Judge Me	2Pac	rap	2
3	Insta	Bizzey	latin	2
4	Being Alive	Hardwell	edm	4
7	Dance?	Night Tempo	r&b	1
11	Idealne Połączenie	Young Igi	rap	2
...	...	...	...	...
492	In My Mind	The Amazons	rock	5
493	Extraños En El Escaparate - Live	Miguel Rios	rock	4
495	Sparkles	Daxten	latin	2
496	Reflections	Jean Juan	edm	4
497	Cryin' Like A Bitch!!	Godsmack	rock	5

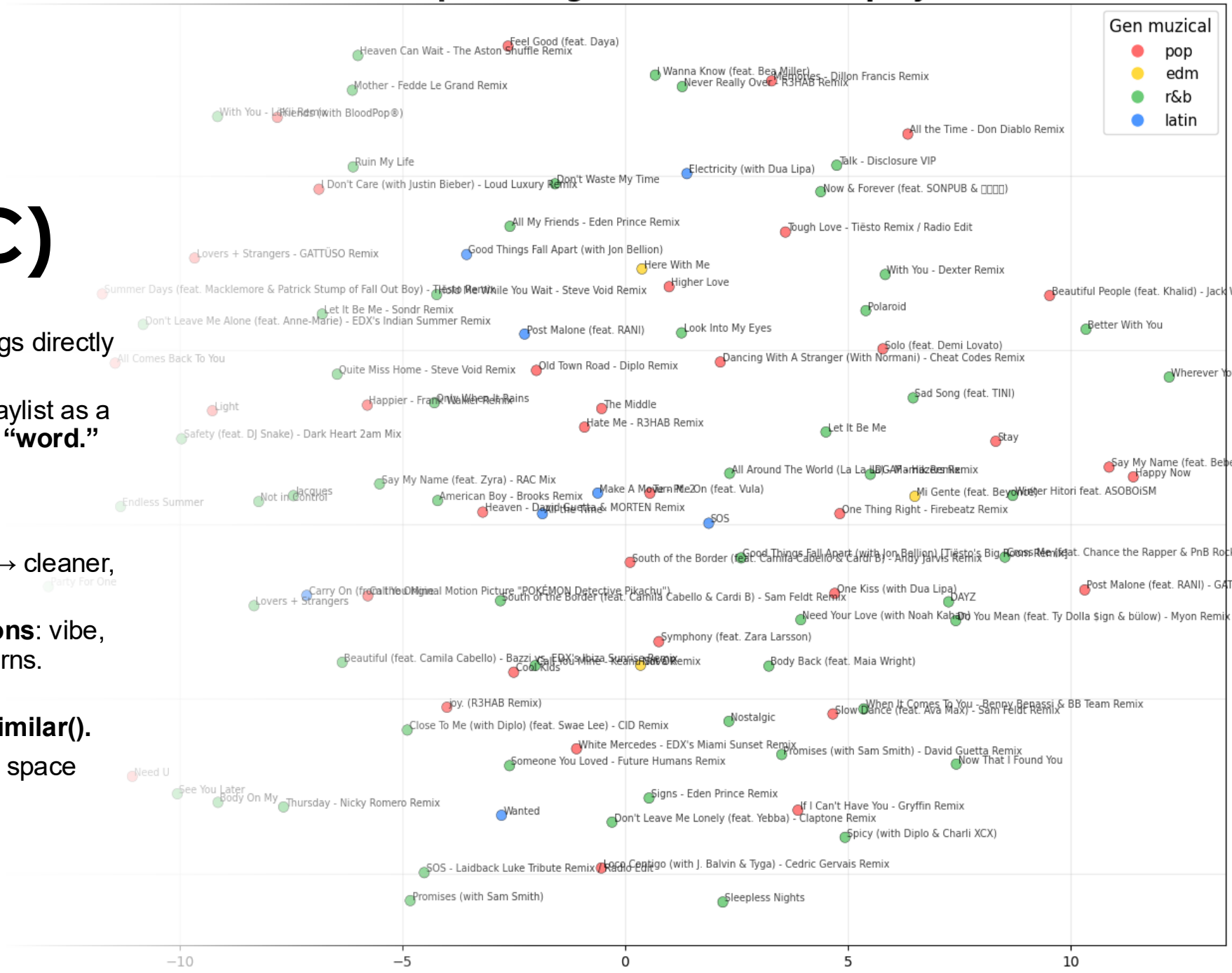
172 rows × 4 columns

SVD Recommendations (new songs):

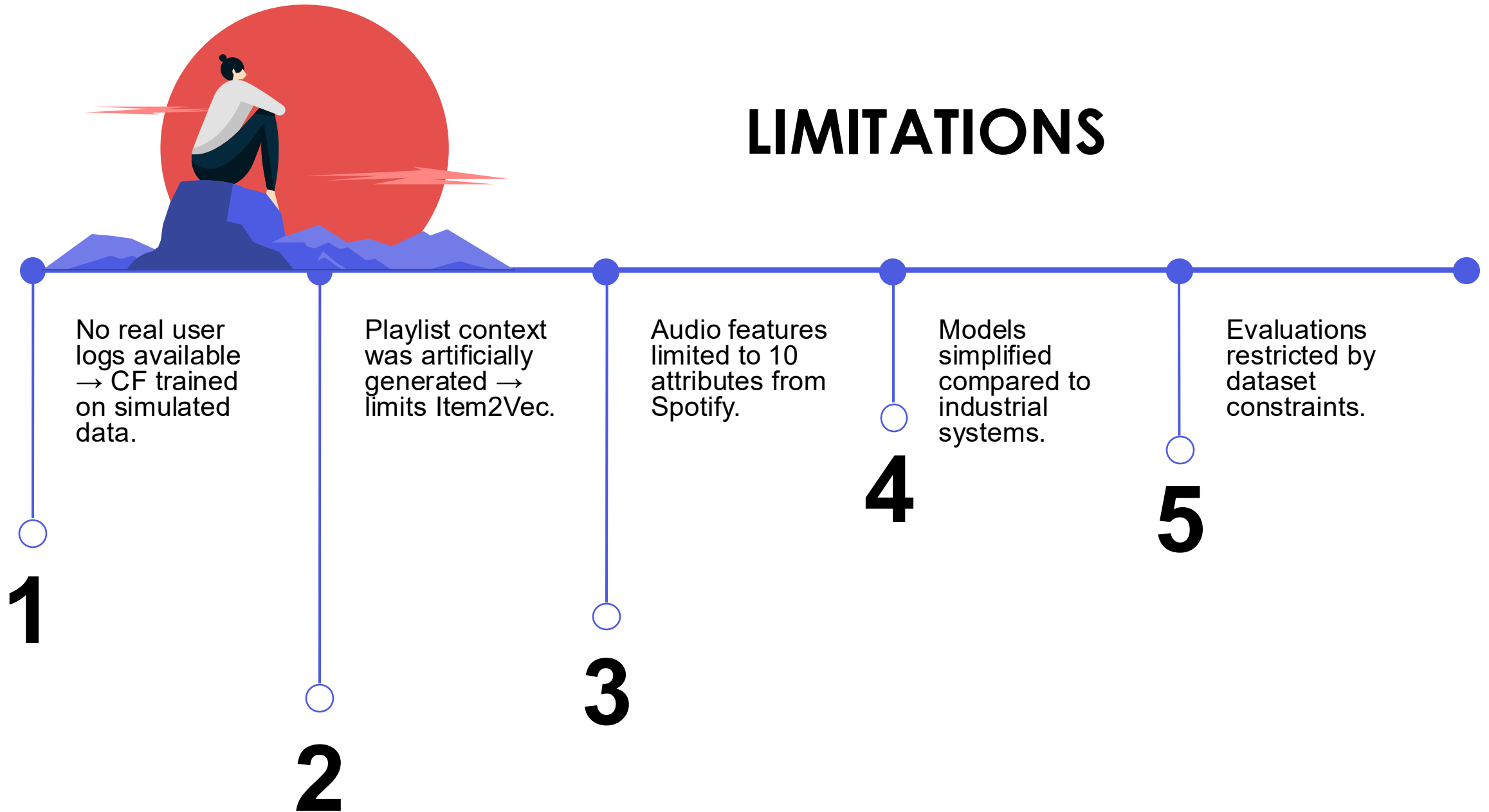
	track_name	track_artist	playlist_genre	predicted_score
107	Wrong Way	Sublime	rock	4.295300
241	Sofia	Alvaro Soler	latin	4.109802
96	Lovesong	The Cure	rock	3.986341
370	The Dark	John Clark	rock	3.898507
391	Here I Go Again 87 - 2017 Remastered Version	Whitesnake	rock	3.874134
487	White Flag	Dido	rock	3.792521
90	(Don't Fear) The Reaper	Blue Öyster Cult	rock	3.768494
94	Bedside Radio - Remastered	Krokus	rock	3.753841
328	All In My Head	Tori Kelly	r&b	3.708414
344	The Climb	Miley Cyrus	pop	3.705294

Semantic Map of Songs (with Names Displayed)

- Gen muzical
- pop
 - edm
 - r&b
 - latin



LIMITATIONS



Andra-Mihaela Andruta & Ioana-Alexandra Tunaru

Select the project stage (model) you want to explore.

Stage 1: Content-Based Filtering

Stage 2: Collaborative Filtering

Stage 3: Neural / Contextual Filtering

Recommendation based on acoustic profile (Cosine Similarity)

Song Name (or part of it) Cryin

Number of Recommendations 5

Get Recommendations

INTERACTIVE UI

-STAGE 1-

Purpose of the UI

- The user interface is built with **Gradio**, providing a lightweight and interactive layer for exploring our recommendation models.

Content-Based Filtering

- This stage allows users to input a song and receive recommendations based on **acoustic similarity**, highlighting interpretable, feature-based matching.

Select the project stage (model) you want to explore.

Stage 1: Content-Based Filtering Stage 2: Collaborative Filtering Stage 3: Neural / Contextual Filtering

Collaborative Filtering (SVD, Co-Listening & Explanations)

1. SVD Recommendations (Latent Factors) 2. Similar Users 3. Co-Listening Recommendations

Co-Listening Recommendations (Users who listened to X also listened to Y)

****Step 1: Verify Song / Artist****

Target Song (Exact name from the table above)

Lovesong

Number of Recommendations

5

↺

1

10

Get Co-Listening Recommendations

Searched song: **Lovesong** by **The Cure** (rock)

Co-Listening Recommendations

Song	Artist	Genre	Co-listen Count
Bad Boy Flow	twoloud	edm	40
I Know What You Want (feat. Flipmode Squad)	Busta Rhymes	rap	40
Voodoo - Original Mix	DVBBS	edm	37
Cosmic Waves - Original Mix	Hell Driver	edm	37
Ginza	J Balvin	latin	37

-STAGE 2-

Collaborative Filtering

- The interface is structured into **three sub-tabs**, each corresponding to a collaborative filtering technique (SVD, Similar Users, Co-Listening)
- Inputs and outputs are explicitly separated, allowing users to follow the recommendation process step by step.

-STAGE 3-

Neural Filtering

- The interface is organized into **three focused sub-tabs**, each showcasing a contextual recommendation scenario
- User inputs (genres, number of steps) directly control the **semantic transition process**
- Results are displayed as an **ordered playlist**, making abstract embedding relationships easy to interpret

 **Music Recommendation System**
Andra-Mihaela Andruta & Ioana-Alexandra Tunaru

Select the project stage (model) you want to explore.

Stage 1: Content-Based Filtering Stage 2: Collaborative Filtering Stage 3: Neural / Contextual Filtering

Context-Based Filtering (Word2Vec)

1. Item2Vec Song Similarity (Mixed Corpus) 2. Genre Compatibility (Structured Corpus) 3. Semantic Transition (Songs)

Transition playlist (Genre A → Genre B) using the Word2Vec semantic space

Starting Genre
rap

Target Genre
rock

Number of Intermediate Steps
5

Semantic Transition: **RAP** → **ROCK** (via 4 songs)

Transition Playlist (Songs & Artists)

Step	Song	Artist
1	Sweet Child O' Mine	Pickin' On Series
2	Tear Me Down (feat. Philip Rustad)	Davaï
3	My Michelle	Guns N' Roses
4	If I Can't Have You - Gryffin Remix	Shawn Mendes

CONCLUSION

- CBF, CF and Item2Vec capture **different dimensions of similarity** (audio, behavioral, contextual).
- **Content-Based Filtering** works well for acoustic similarity but lacks personalization.
- Collaborative Filtering (**SVD**) provides stronger **personalized recommendations** through latent factors.
- **Co-listening patterns** are effective for identifying culturally similar tracks.
- **Item2Vec** learns deeper **contextual relationships** between songs from playlist structure.
- **PCA and t-SNE** clearly show how each model organizes the musical space.
- Combining multiple methods creates a **more robust recommender system**.





THANK YOU!

0:16

Additional details and the complete implementation can be found
on our GitHub repository.

3:37

